Protect Privacy Through Forgetting: Machine Unlearning for Vision Models

Saurabh Chatterjee (NS26Z334) Reva Bharara (NS26Z212)
CS6886 Systems Engineering for Deep Learning, IIT Madras
Code: https://github.com/chatterjeesaurabh/Machine-Unlearning-for-Vision-Models

1 Introduction

Privacy laws (GDPR's right to erasure, India's DPDP Act 2023) and data-quality problems require removing the influence of selected training data — the *forget set* D_f — from a trained model, while keeping its behaviour on the *retain set* D_r. Retraining from scratch on D_r (the gold standard θ_r) costs a full training run per request. *Approximate unlearning* updates the original model θ_o for a few epochs instead, and must be judged by how closely it matches θ_r: in accuracy, in privacy (membership inference) and inside the learned representation. We study two vision settings on ResNet-18/CIFAR-10: **class forgetting** (S1: all 4,500 airplanes, e.g. removing an identity) and **subset forgetting** (S2: a random 10/30/50% of cat, dog and truck, e.g. removing individual users' images), where θ_r still recognises 89–92% of the forgotten images.

2 Prior work

Exact unlearning retrains, but limits the cost: SISA [1] trains an ensemble on disjoint shards so that a deletion retrains only one shard; Amnesiac Machine Learning [2] stores the parameter updates of each batch and subtracts those that involved deleted data. *Approximate* unlearning modifies a single trained model: Fisher forgetting [3] adds noise to the weights according to the Fisher information of the retain data, Thudi et al. [4] analyse how far SGD unlearning is from retraining, and Jia et al. [5] show that sparser (pruned) models are easier to unlearn. The two fine-tuning **baselines** we build on are **Gradient Ascent (GA)**, which maximises the forget loss, $\theta \leftarrow \theta + \eta \nabla L(\theta; D_f)$ [4], and **GA + Descent**, which balances it with the retain loss, $\min_\theta \beta L(D_r) - (1-\beta) L(D_f)$ [9]. Evaluation follows membership inference attacks [6] and recent concerns that forgetting may be only output-deep [7] or reversible under input perturbation [8].

3 Our exploration and implementation

Beyond the two baselines we explore four fine-tuning strategies, each designed and tuned on our own protocol, and a weight-selection add-on:

1. **Teacher-Student unlearning** (building on distillation-based unlearning [9]): a frozen copy of θ_o is the teacher; the student alternates *max-steps* (maximise $\mathrm{KL}(p_T \| p_S)$ on D_f) and *min-steps* ($\alpha\,\mathrm{KL} + \gamma\,\mathrm{CE}$ on D_r). We found that soft labels at temperature 4 re-teach the forgotten class (5% airplane probability on retain images) and that, in S2, the method is bistable: it forgets nothing or collapses. We use a small KL weight and scale the forget batch with $|D_f|$ so that the number of forget updates stays constant.
2. **Random Relabeling** [11]: fine-tune on forget images with random wrong labels interleaved with retain passes.
3. **Boundary (nearest-class) Relabeling** (building on [10]): relabel each forget image to the Original's second most likely class (instead of an adversarial search, which gave inconsistent labels such as frog for airplanes), add a retain pass and *mixed BN batches* (below).
4. **GA with mixed BN batches and iteration-level early stopping** at the Retrain forget accuracy.
5. **Gradient based Weight Saliency** (WS) [11]: update only the 50% of weights with the largest forget-loss gradient; applied on top of every method.

Key observation — BatchNorm statistics fake forgetting. Methods trained on forget-only batches overwrite the BN running statistics within ≈30 steps. In S1, plain GA changes the weights by only 2.0% but the BN running means by 24%; re-estimating only the BN statistics on retain data brings forget accuracy back from 5.3% to 92% (Fig. 1, bottom). The "forgetting" was normalisation drift that any BN recalibration (e.g. during deployment) undoes. Our fix, **mixed BN batches**, concatenates a retain batch to every forget forward pass while the loss stays on the forget part; forgetting then comes from the weights.

Protocol and systems. 45k/5k/10k train/validation/test split; Original: 182 epochs, 94.82% test accuracy; a Retrain reference for every scenario and seed (3 seeds). Hyperparameters were selected with a validation-only gap, never with test data. Metric: avg_gap $= \frac{1}{4}\sum|\Delta|$ over forget, retain and test accuracy and MIA accuracy w.r.t. θ_r (0 = identical to retraining). MIA: logistic regression on per-sample losses, forget vs. unseen same-class test images, 5-fold CV (Retrain ≈50%). The dataset is held on the GPU with GPU augmentation and bf16 (a 182-epoch retraining takes 6–8 min on an A100); a GPU lock, OOM-retry and a parallel runner (15 jobs on 2 GPUs) executed 1,845 recorded runs.

4 Evaluation against the baselines

Table 1: avg_gap to Retrain (3 seeds, lower is better) and S1 MIA (Retrain 51.0%). Baselines: GA, GA + Descent.

Method	S1	S2-10	S2-30	S2-50	MIA S1
No unlearning (Original)	29.05	4.47	4.75	5.26	57.8
Baseline: GA	4.58	7.20	5.55	8.78	51.2
Baseline: GA + Descent	1.34	**3.52**	4.15	3.80	49.3
GA + mixed BN (ours)	1.07	4.80	4.40	4.18	51.8
Teacher-Student	0.60	3.81	3.34	3.62	52.6
Random Relabeling + WS	1.17	4.36	**3.10**	**2.58**	55.5
Boundary Relabeling	**0.16**	11.22	9.92	8.98	**51.0**

Accuracy and privacy (Table 1, Fig. 1). In class forgetting all explored methods outperform both baselines; Boundary Relabeling with mixed BN matches retraining (forget 0%, retain 100%, MIA 51.03 vs. 51.02; avg_gap 0.16) in 28 s instead of 402 s, and Teacher-Student is second (0.60). Mixed BN alone brings plain GA from 4.58 to 1.07. Random Relabeling reaches 0% forget accuracy but the highest MIA (55.5%): random labels make forget images recognisably high-loss. In subset forgetting the best methods reduce the gap of doing nothing (4.5–5.3) to 2.6–3.5: Random Relabeling + WS on S2-30/50, GA + Descent on S2-10, while Boundary Relabeling over-forgets and damages same-class retain images. Every method is 14–108× faster than retraining.

Entanglement decides difficulty in subset forgetting (Table 2). For every method, forget accuracy rises from the least to the most entangled third, and pushing entangled im-

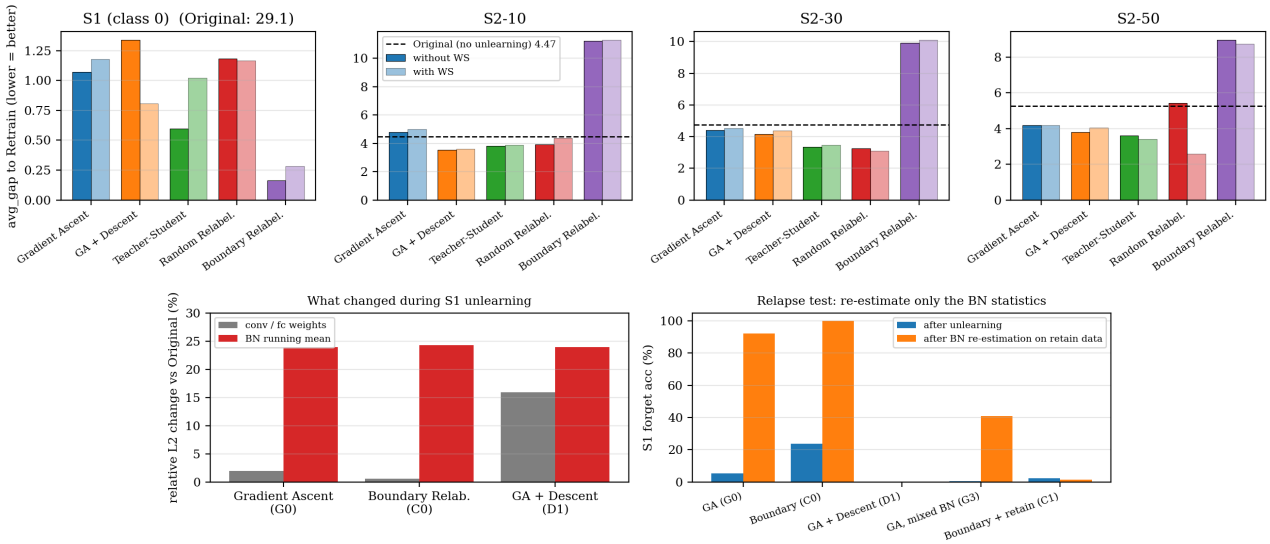


Figure 1: Top: avg_gap to Retrain of every method (solid: without WS, light: with WS; dashed: doing nothing). Bottom: BatchNorm statistics in S1 — with forget-only batches the weights barely change while the BN running means change by ≈24% (left); re-estimating only the BN statistics brings the "forgotten" class back unless the method uses mixed BN batches or a retain pass (right).

Table 2: Entanglement trend, S2-30: forget accuracy (%) after unlearning each third of the forget set alone (low = far from retain images in feature space, high = close).

Model	low	medium	high	retain acc (high)
Retrain	79.0	94.7	99.6	100.0
Gradient Ascent	77.7	93.2	99.2	99.2
GA + Descent	82.7	94.9	94.8	97.8
Teacher-Student	64.1	93.0	98.4	98.1
Random Relabeling	89.5	96.9	98.9	97.1
Boundary Relabeling	52.1	74.6	68.7	89.2

ages down costs retain accuracy (Boundary: 89.2%). Low-entanglement images are the memorised ones (Retrain recognises only 79%). In class forgetting, by contrast, every third is forgotten ($\leq$2.8%) and the forgetting spreads to the unused airplanes: a class is one shared concept. Unlearning the thirds sequentially (low $\rightarrow$ high) helps only Random Relabeling (S2-50 5.43 $\rightarrow$ 2.55); ascent-based methods accumulate damage.

Output-level vs. representation-level forgetting (S1). A linear probe on frozen features recovers the forgotten class at 82–94% for all methods, but also at 87.7% for Retrain, so probe accuracy alone does not prove residual memory. Nearest-centroid accuracy separates the methods: GA, GA + Descent and Teacher-Student disperse the airplane cluster (30–46%, Retrain 41%), while both relabeling methods keep it intact (75–82%): their forgetting lives in the classifier head.

Weight Saliency improves methods that damage the retain set (Boundary main setting 6.92 $\rightarrow$ 4.03, GA + Descent 1.34 $\rightarrow$ 0.81, Random Relabeling S2-50 5.43 $\rightarrow$ 2.58), beats a random mask of the same density for every method, and stabilises Teacher-Student, but cannot protect BN statistics (they are buffers, not weights).

Robustness. Weight-only INT8/6/4 quantization never restores forgotten knowledge (max. +0.42 points). In S2, INT4 lowers forget accuracy (by up to 8 points) together with retain and test accuracy, because unlearning leaves forget-class images with low margins (16.5% of correct forget images for GA vs. 5.3% for Retrain). Under Gaussian input noise no S1 model recognises the forgotten class.

Conclusion. Fine-tuning-based unlearning reaches retraining quality for class forgetting once BN statistics, label choice and schedules are handled; evaluations should include a BN re-estimation check and representation-level metrics. Subset forgetting is bounded by generalisation; its remaining gap is privacy on memorised, low-entanglement samples.

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